CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

6.1 Stress & Strain

Syllabus Section: 6. Deformation of Solids **Paper Type:** Multiple Choice (Paper 1) **Total:** 25 Questions (25 Marks)

Question 1 Easy [1 mark]

A student wanted to measure the Young modulus of a wire, to do this they needed to take a number of measurements.

In which row could the measurement **not** be made **directly** with the listed apparatus?

	Measurement	Apparatus
A	extension of wire	vernier scale
B	area of cross-section of wire	micrometer screw gauge
C	original length of wire	metre rule
D	mass of load applied to a wire	electronic balance

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The area of the cross-section of the wire = πr^2
- A micrometer can measure the diameter of the wire with a resolution of 0.001 mm
- Therefore, a micrometer cannot directly measure the cross-sectional area of the wire

A is incorrect because the vernier scale will measure the extension of the wire

C is incorrect because the length of wire could be measured directly by a metre rule

D is incorrect because an electronic balance will measure the mass of the load directly

Question 2 Easy [1 mark]

What is the unit of the Young modulus?

A N m^{-2}

B N m

C N m^{-1}

D N m^{2}

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The Young modulus is given by $E = \frac{stress}{strain} = \frac{\sigma}{\varepsilon}$
  - Stress is equal to $\sigma = \frac{force}{area} = \frac{F}{A} = \mathrm{Nm^{2}}^{-1} = \mathrm{N\ \ m^{-2}}$
  - Strain is equal to $\varepsilon = \frac{extension}{original\ \ length} = \frac{\Delta L}{L} = \mathrm{mm^{-1}} = 1$
- Strain is dimensionless, so, the units for the Young modulus are N m^{-2}

Question 3 Easy [1 mark]

A material is extended elastically within the limit of proportionality. What is equal to the Young modulus of the material?

A gradient of the stress-strain graph

B area under the force-extension graph

C area under the stress-strain graph

D gradient of the force-extension graph

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The Young modulus is given by $= E = \frac{stress}{strain} = \frac{\sigma}{\varepsilon}$
  - Stress is equal to $\sigma = \frac{force}{area} = \frac{F}{A}$
  - Strain is equal to $\varepsilon = \frac{extension}{original\ \ length} = \frac{\Delta L}{L}$
- The Young modulus is equal to the gradient of the stress-strain graph

B is incorrect because the area under a force-extension graph is the energy stored in the spring

C is incorrect because the area under a stress-strain graph is the toughness

D is incorrect because the gradient of a force-extension graph is the spring constant, *k*

Question 4 Easy [1 mark]

An experiment was carried out to determine the Young modulus of steel using a steel wire.

The uncertainties in the measurements are given below.

Measurement	Uncertainty
load on wire	±2%
length of wire	±0.2%
diameter of wire	±1.5%
extension	±1%

What is the percentage uncertainty in the Young modulus?

A 1.3%

B 1.8%

C 4.7%

D 6.2%

Mark Scheme & Explanation:

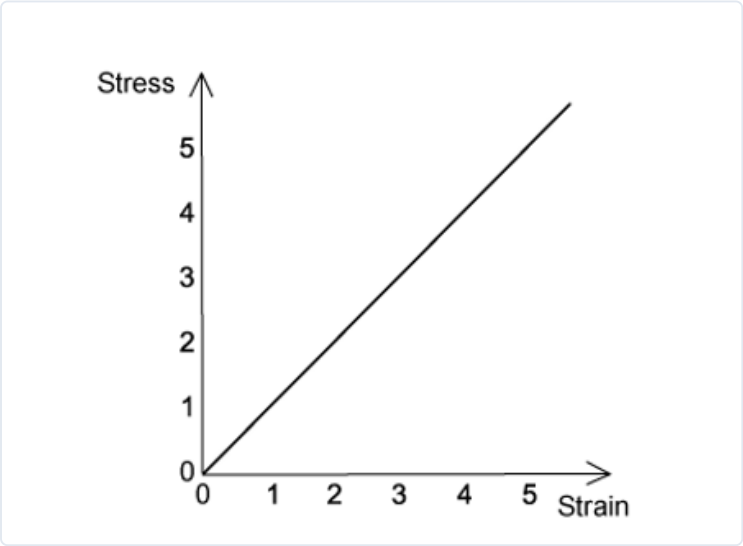
The correct answer is D because:

- The equation for Young's modulus E is:
  - $E = \frac{F}{A} \times \frac{L}{x}$
  - where F = force (load), A = area, L = length, and x = extension
- For a wire, the cross-sectional area A is:
  - $A = \pi r^2 = \pi \frac{d}{2}^2 = \frac{\pi d^2}{4}$
  - where d = diameter
- To find the percentage uncertainty in E, we add the percentage uncertainties in the measurements
  - % uncertainty in F = 2%
  - % uncertainty in L = 0.2%
  - % uncertainty in d = 1.5%

    - % uncertainty in x = 1%
- We can ignore any constants, such as $\frac{\pi}{4}$
- The uncertainty in the diameter must be multiplied by 2 since it is squared
    - % uncertainty in E = 2% + 0.2% + (2 × 1.5%) + 1%
    - % uncertainty in E = 6.2%

Question 5 Easy [1 mark]

A glass rod was put under stress and strain until it broke. The graph of this is shown below.



Which of the following statements is correct about the glass rod?

A the glass shows plastic deformation

B when the cross-sectional area of the rod is doubled, the ultimate tensile stress of the rod is halved

C Hooke's law is obeyed for all values of stress up to the breaking point

D the glass is ductile

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Hooke's law states that the strain in a solid is proportional to the applied stress within the elastic limit
- The graph is a straight line going through the origin, showing proportionality
- Therefore, Hooke's law is obeyed for all values of stress up to the breaking point

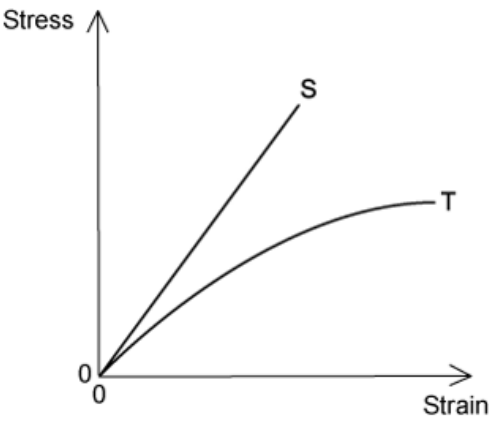
A is incorrect because plastic deformation occurs after the limit of proportionality and is after the elastic limit, glass is brittle and does not show plastic deformation

B is incorrect because ultimate tensile stress is a material property independent of cross-sectional area — it is the stress at fracture, so it remains constant regardless of geometry. Doubling the area would double the breaking force, not the ultimate tensile stress

D is incorrect because glass is brittle and does not show plastic deformation as a ductile material would

Question 6 Easy [1 mark]

Two wires S and T, with the same initial dimensions, were put under stress and strain, the graph shows the results up to their breaking points.



Which statement is **not** correct?

A material S has a larger ultimate tensile stress

B material S extends elastically

C material S extends more than material T when loaded with the same force

D material S is brittle

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The Young modulus is given by $E = \frac{stress}{strain} = \frac{\sigma}{\varepsilon}$
  - Stress is equal to $\sigma = \frac{force}{area} = \frac{F}{A}$
  - Strain is equal to $\varepsilon = \frac{extension}{original\ \ length} = \frac{\Delta L}{L}$
- The strain is determined by the extension of the material and T has a larger strain than S

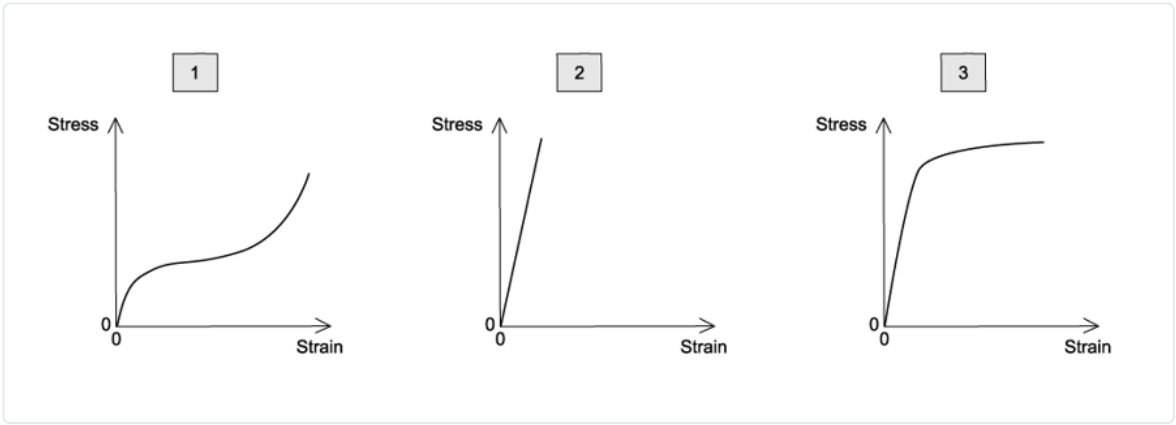
A is incorrect because the ultimate tensile strength is the maximum stress a material can withstand before breaking; this is higher for S than T

B is incorrect because material S shows proportionality and extends elastically

D is incorrect because material S shows no plastic deformation and is brittle

Question 7 Easy [1 mark]

The graphs below show the stress-strain graphs of three materials. The graphs do not have the same scales.



The three materials are copper, rubber and glass.

Which materials are represented by the graphs?

	1	2	3
A	rubber	copper	glass
B	rubber	glass	copper
C	glass	copper	rubber
D	copper	glass	rubber

A

B

C

D

Mark Scheme & Explanation:

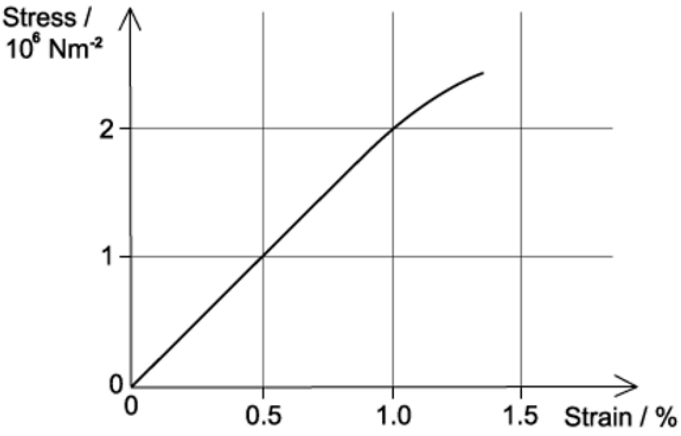
*The correct answer is **B** because:*

- Material 1 shows an unusual shape stress-strain graph associated with a polymer, such as rubber
  - Rubber deforms elastically up to a certain point as the polymers chains, which were previously curled up, uncurl
  - After its yield point is reached it will no longer return to its original shape as it undergoes hysteresis
- Material 2 shows a brittle material that shows no plastic deformation, such as glass

    - Glass is described as brittle as it does not extend very much and when subjected to stress, it breaks with little elastic deformation
- Material 3 shows a ductile material that shows plastic deformation, such as a metal wire
    - A metal wire tends to be described as ductile as it will experience a large amount of extension when subjected to stress, especially beyond the elastic limit

Question 8 Easy [1 mark]

The stress-strain graph for bone is shown below.



What is the Young modulus of bone?

A 2×10^6 N m^{-2}

B 2×10^8 N m^{-2}

C 1×10^6 N m^{-2}

D 1×10^8 N m^{-2}

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The Young modulus is given by $E = \frac{stress}{strain} = \frac{\sigma}{\epsilon}$
  - Stress is equal to $\sigma = \frac{force}{area} = \frac{F}{A}$
  - Strain is equal to $\varepsilon = \frac{extension}{original\ \ length} = \frac{\Delta L}{L}$
- From the graph:
  - Stress = 2×10^6 N m^{-2}
  - Strain $= 1\,\% = \frac{1}{100} = 0.01$
- Young modulus $= \frac{2 \times 10^6}{0.01} = 2 \times 10^8$ N m^{-2}

Exam Tip: a common error students make is that they tend to forget to check the units on the graph's axes. It would be an easy mistake to make to take the percentage strain from the graph and not convert it back to the ratio extension/original length.

Question 9 Easy [1 mark]

A student wanted to determine the spring constant of a spring. They did this by using different loads to extend the spring by varying amounts.

To find the spring constant, which quantities are required?

A the elastic limit, extensions and the length of the spring

B the loads and the extensions of the spring

C the elastic limit and the loads

D the loads and the length of the spring

Mark Scheme & Explanation:

*The correct answer is **B** because:*

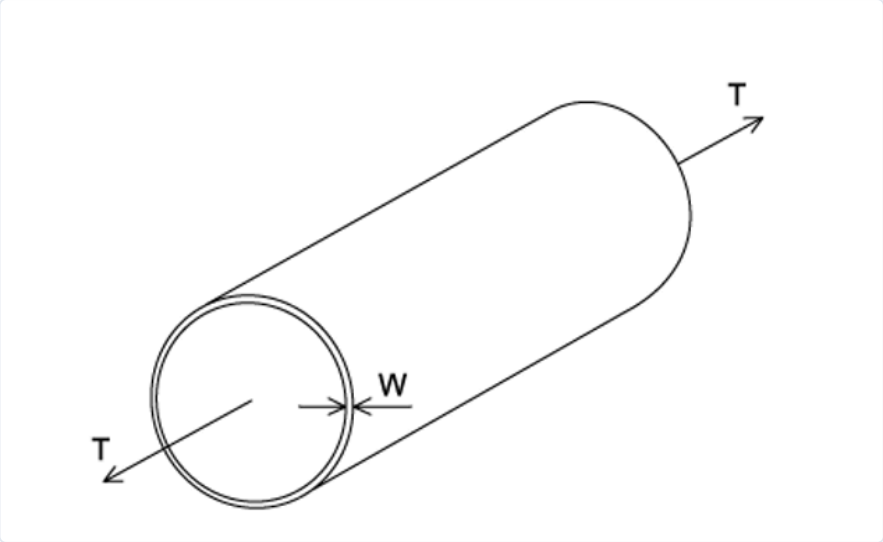
- Hooke's law states that the strain in a solid is proportional to the applied stress within the elastic limit
- Hooke's law states that $F = kx$, where *k* is the spring constant, *F* is the force and *e* is the extension
    - So, to calculate the spring constant $k = \frac{F}{x}$
- To do this, the force (load) and the extension of the spring are required

A & D are incorrect because the elastic limit is the point that the material will no longer obey Hooke's law, it is calculated using the extension not the length of the material

C is incorrect because the elastic limit is the point that the material will no longer obey Hooke's law

Question 10 Medium [1 mark]

A metal tube of diameter d with a thin wall thickness W is shown in the diagram. The thickness of the wall is small when compared to the diameter of the tube.



A force of magnitude T is applied parallel to the axis of the tube, putting the tube under tension. It is proposed that making the wall thicker would reduce the stress on the tube.

If the tube diameter and the tension remain the same, which wall thickness would half the stress?

A $\frac{W}{2}$

B $\sqrt{2}W$

C $2W$

D $4W$

Mark Scheme & Explanation:

The correct answer is C because:

- The metal tube is hollow with a wall of width W
- It is the area of the tube, not the hollow space, that is taken into account
  - Stress $= \frac{\text{force}}{\text{area}} = \frac{F}{A}$
- Area of the wall = circumference × width $= \pi dW$
- As diameter d is constant, the area of the wall is proportional to the width W

- The tension force T is kept constant, so the stress is inversely proportional to the area
  - Stress $\propto \frac{1}{\pi dW}$
- So, to halve the stress, the thickness W of the wall would need to be doubled
  - Therefore, the new wall thickness would be $2W$

The expression for area, $A = \pi dW$, is an approximation which is valid since the walls of the tube are said to be very thin. The cross-sectional area of a hollow cylindrical tube is the difference between the areas of the outer radius and the inner radius:

$$A = \pi\frac{d}{2}^2 - \pi\frac{d - 2W}{2}^2$$

$$A = \frac{\pi}{4}d^2 - d - 2W^2$$

$$A = \frac{\pi}{4}d^2 - d^2 + 4W^2 - 4dW$$

Since $W \ll d$, this means the $4W^2$ term can be ignored, leading to:

$$A \approx \frac{\pi}{4}d^2 - d^2 + 4dW \approx \frac{\pi}{4}4dW$$

$$A \approx \pi dW$$

Question 11 Medium [1 mark]

A steel metal wire has the following properties:

diameter = 5.0×10^{-4} m

Young modulus = 2.0×10^{11} Pa

tension = 20 N

The string snaps and contracts elastically.

By what percentage does a length *l* of a piece of the string contract?

A $1.3 \times 10^{-4}\,\%$

B $5.1 \times 10^{-4}\,\%$

C $1.3 \times 10^{-2}\,\%$

D $5.1 \times 10^{-2}\,\%$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The strain of the metal wire is equal to $\varepsilon = \frac{extension}{original\ \ length} = \frac{\Delta L}{L}$
- The percentage of the length that contracts = strain × 100%
  - So, the percentage of the length that contracts $= \frac{\Delta L}{L} \times 100\%$
- The Young modulus is given by $E = \frac{stress}{strain} = \frac{\sigma}{\varepsilon}$
  - Stress is equal to $\sigma = \frac{force}{area} = \frac{F}{A}$
- The cross-sectional area of the wire $A = \frac{\pi d^2}{4}$
- So, strain is equal to
  - $\varepsilon = \frac{\sigma}{E} = \frac{1}{E}\frac{F}{A} = \frac{F}{E}\frac{4}{\pi d^2}$
  - $\varepsilon = \frac{20}{2.0 \times 10^{11}} \times \frac{4}{\pi 5.0 \times 10^{-4\,2}} = 5.09 \times 10^{-4}$
- So, as a percentage this is:
  - Strain × 100% = 5.09×10^{-4} × 100% = 5.09×10^{-2} %

- Therefore, the percentage of the length that contracts is equal to 5.1×10^{-2} %

Question 12 Medium [1 mark]

The Young modulus of a metal wire is dependent on which property?

A spring constant

B ductility

C ultimate tensile stress

D elastic limit

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The Young modulus is given by $E = \frac{stress}{strain} = \frac{\sigma}{\varepsilon}$
  - Stress is equal to $\sigma = \frac{force}{area} = \frac{F}{A}$
  - Strain is equal to $\varepsilon = \frac{extension}{original\ \ length} = \frac{\Delta L}{L}$
- Therefore, Young modulus $E = \frac{\frac{F}{A}}{\frac{x}{L}} = \frac{FL}{Ax}$
- Hooke's law states $F = ke$, where k is the spring constant
- A metal wire will obey Hooke's law up to a point, the spring constant depends on the force, and this can be found in the equation for the Young modulus:
  - $E = \frac{keL}{Ax} = \frac{kL}{A}$
- So, Young modulus E depends on the **spring constant** (and original length L and cross-sectional area A)

B is incorrect because not all materials are ductile, but all materials can have a Young modulus

C is incorrect because the ultimate tensile strength is the maximum stress a material can withstand before breaking; the Young modulus does not depend on this

D is incorrect because the elastic limit is beyond the limit of proportionality; the Young modulus does not depend on this

Question 13 Medium [1 mark]

The Young modulus *E* and the force per unit extension *k* describes the behaviour of a wire under tensile stress.

For a wire of length *L* and cross-sectional area *A*, what is the relationship between *E* and *k* ?

A $E = \frac{A}{kL}$

B $E = \frac{kA}{L}$

C $E = \frac{kL}{A}$

D $E = \frac{L}{kA}$

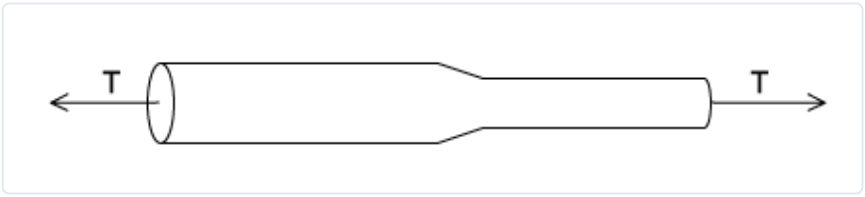
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The Young modulus is given by $E = \frac{stress}{strain} = \frac{\sigma}{\varepsilon}$
  - Stress is equal to $\sigma = \frac{force}{area} = \frac{F}{A}$
  - Strain is equal to $\varepsilon = \frac{extension}{original\ \ length} = \frac{\Delta L}{L}$
- Therefore, Young modulus $E = \frac{\frac{F}{A}}{\frac{x}{L}} = \frac{FL}{Ax}$
- Hooke's law states that $F = kx$, so $k = \frac{F}{x}$
- Therefore, Young modulus is equal to
  - $E = \frac{FL}{Ae} = \frac{L}{A}\frac{F}{e} = \frac{kL}{A}$

Question 14 Medium [1 mark]

A steel bar, shown in the diagram, has a circular cross-section that is under tension *T*.



The diameter of the thinner portion is half the wider portion.

What is the value of $\frac{stress\ \ on\ \ the\ \ wide\ \ portion}{stress\ \ on\ \ the\ \ narrow\ \ portion}$?

A 4.0

B 2.0

C 0.50

D 0.25

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Stress is equal to $= \frac{force}{area} = \frac{F}{A}$
- The steel tube is thicker at one end by double that of the thinner end
- The tension force is kept constant, so the stress is inversely proportional to the area
- The ratio of the stress in the two sections would be 0.25

Example of working:

narrow portion	wide portion
Diameter = 0.2 m, force = 10 Area $= \pi r^2 = \pi \times 0.1^2 = 0.0314\ \text{m}^2$ Stress $= \frac{force}{area} = \frac{F}{A}$ Stress $= \frac{10}{0.0314} = 318.5\ \text{N m}^{-2}$	Diameter 0.4 m, force 10 N Area $= \pi r^2 = \pi \times 0.2^2 = 0.1256\ \text{m}^2$ Stress $= \frac{force}{area} = \frac{F}{A}$ Stress $= \frac{10}{0.1256} = 79.6\ \text{N m}^{-2}$

- So, the value of $\frac{stress\ \ on\ \ the\ \ wide\ \ portion}{stress\ \ on\ \ the\ \ narrow\ \ portion} = \frac{79.6}{318.5} = 0.25$

Question 15 Medium [1 mark]

A student wanted to find the young modulus of a wire. The equation for the young modulus *E* is:

$$E = \frac{4Fl}{\pi d^2 x}$$

The student extended the wire with a known force and made a series of measurements.

Which measurement has the largest effect on the uncertainty in the value of the calculated Young modulus?

	Measurement	Symbol	Value
A	length of wire before force applied	l	2.043 ± 0.002 m
B	force applied	F	19.62 ± 0.01 N
C	extension of wire with force applied	x	5.2 ± 0.2 mm
D	diameter of wire	d	0.54 ± 0.02 mm

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

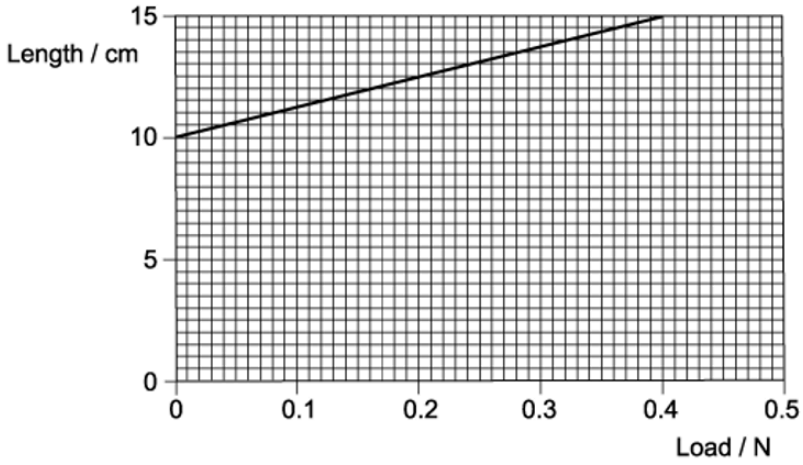
- The Young modulus is given by $E = \frac{stress}{strain} = \frac{\sigma}{\epsilon}$
  - Stress is equal to $\sigma = \frac{force}{area} = \frac{F}{A}$
  - Strain is equal to $\varepsilon = \frac{extension}{original\ \ length} = \frac{\Delta l}{l}$
- The cross-sectional area of the wire $A = \frac{\pi d^2}{4}$
- So, the Young Modulus for a wire is given by
  - $E = \frac{4Fl}{\pi d^2 x}$
- The percentage uncertainty in the diameter needs to be doubled as it is squared in the equation
  - $\frac{\Delta E}{E} = \frac{\Delta F}{F} + \frac{\Delta l}{l} + 2\frac{\Delta d}{d} + \frac{\Delta x}{x}$
    - $\frac{\Delta F}{F} = \frac{0.01}{19.62} = 0.00051$

- $\frac{\Delta l}{l} = \frac{0.002}{2.043} = 0.00098$
- $2\frac{\Delta d}{d} = 2\frac{0.02}{0.54} = 0.07407$
- $\frac{\Delta x}{x} = \frac{0.2}{5.2} = 0.03846$

- Therefore, the diameter has the largest effect on the uncertainty of the Young modulus

Question 16 Medium [1 mark]

A spring was stretched with increasing load. The graph of the results is shown below.



What is the spring constant?

A 0.080 N m^{-1}

B 0.13 N m^{-1}

C 2.7 N m^{-1}

D 8.0 N m^{-1}

Mark Scheme & Explanation:

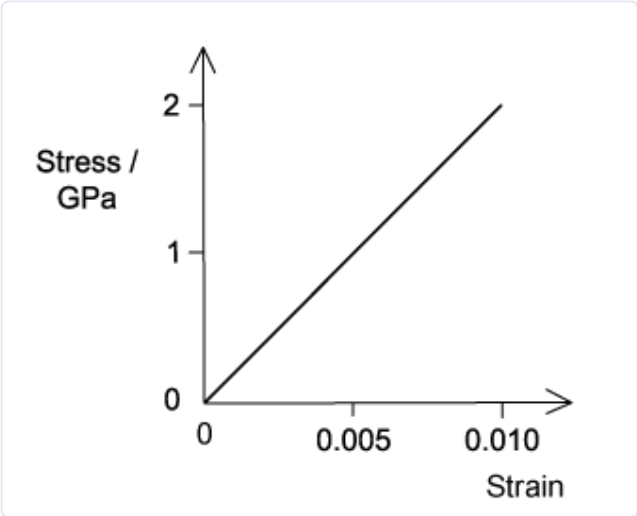
*The correct answer is **D** because:*

- Hooke's law states $F = kx$, where *k* is the spring constant
- The gradient of a force-extension graph is the spring constant
  - Load = 0.4 N
- Extension of the spring *x* = 15 – 10 = 5 cm = 0.05 m
  - $k = \frac{F}{x}$
  - $k = \frac{0.4}{0.05} = 8.0 \text{ N m}^{-1}$
- Therefore, the spring constant is equal to 8.0 N m^{-1}

> Your answer will be more accurate if you use the entire line to calculate the gradient.

Question 17 Medium [1 mark]

The graph below shows the stress-strain for a metal.



When the strain on the rod is 0.010, what is the strain energy per unit volume of a rod made from this metal?

A 10 kJ m^{-3}

B 10 MJ m^{-3}

C 100 kJ m^{-3}

D 1.0 MJ m^{-3}

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- When a material behaves elastically, the work done is stored as energy per unit volume – **strain energy**
- The strain energy is derived from the area under the stress-strain graph
  - Area under the graph $= \frac{1}{2} \times base \times height$
  - Strain energy $= \frac{1}{2} \times 0.010 \times 2 \times 10^{9} = 1 \times 10^{7}$ J $m^{-3} = 10$ M J m^{-3}
- Therefore, the strain energy is equal to 10 MJ m^{-3}

Question 18 Medium [1 mark]

A steel spring was stretched to a length of 55 cm when a 25 N weight was hung from it. The spring has a spring constant of 150 N m^{-1}.

What was the original length of the spring?

A 0.61 m

B 0.72 m

C 0.38 m

D 0.49 m

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Hooke's law states that $F = kx$, where k is the spring constant
- Let the original length = l
- When the force F = 25 N, the extension $x = 55 \quad cm - l = 0.55 \quad m - l$
  - $F = k \times 0.55 - l$
  - $\frac{F}{k} = 0.55 - l$
  - $l = 0.55 - \frac{F}{k} = 0.55 - \frac{25}{150} = 0.383 \quad \text{m}$
- Therefore, the original length of the spring is 0.38 m

Question 19 Medium [1 mark]

A steel cable used in a construction crane has a Young's modulus of 2.1×10^{9} Pa. The cable has a cross-sectional area of 0.05 mm^2 and is subjected to a tensile force of 210 N. The cable obeys Hooke's law.

What is the stress in the steel cable?

A 4.2×10^{3} Pa

B 4.2×10^{6} Pa

C 4.2×10^{9} Pa

D 4.2×10^{12} Pa

Mark Scheme & Explanation:

The correct answer is C

- The stress is given by

$$\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$$

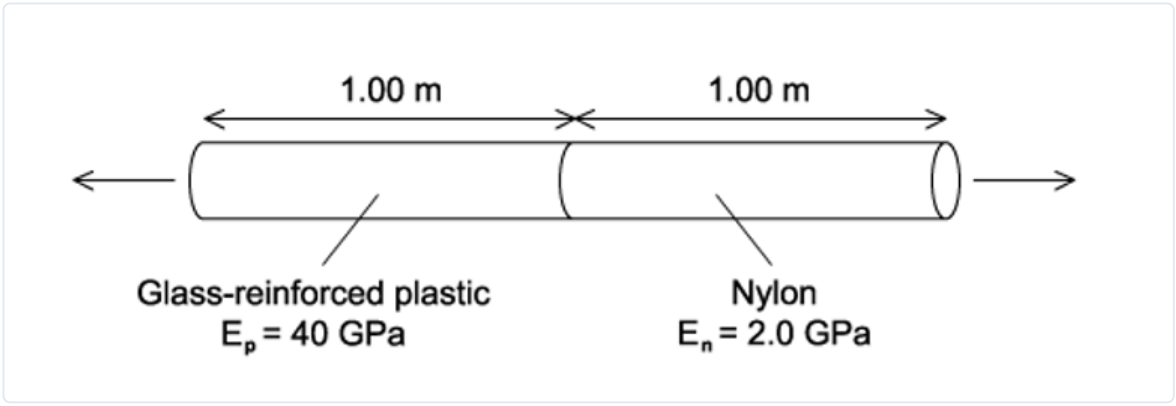
- Where:
  - Force, $F = 210$ N
  - Area, $A = 0.05$ mm^2 $= 0.05 \times 10^{-6}$ m^2

$$\sigma = \frac{210}{0.05 \times 10^{-6}} = 4.2 \times 10^{9} \text{ Pa}$$

- Therefore, the stress in the steel cable is 4.2×10^{9} Pa

Question 20 Hard [1 mark]

A glass-reinforced plastic rod and a nylon rod attached, as shown, was used to make a composite rod.



Each rod has a length of 1.00 m and the same cross-sectional area. The Young modulus E_n of the nylon is 2.0 GPa, and the Young modulus E_p of the plastic is 40 GPa

The composite rod will break when its total extension reaches 3.0 mm.

What is the greatest tensile stress that can be applied to the composite rod before it breaks?

A 5.7×10^{6} Pa

B 5.7×10^{9} Pa

C 7.1×10^{-14} Pa

D 7.1×10^{-2} Pa

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The Young modulus is given by $E = \frac{stress}{strain} = \frac{\sigma}{\varepsilon}$
  - Stress is equal to $\sigma = \frac{force}{area} = \frac{F}{A}$
  - Strain is equal to $\varepsilon = \frac{extension}{original\ \ length} = \frac{\Delta L}{L}$
- Tensile stress is the force applied per cross-sectional area
- In both materials the same force is applied, the materials have different extensions for this force
- The total extension can't be equal to 3.0 mm as the rod would then have broken
  - $E_p = \sigma \times \frac{1.0}{40 \times 10^9} = \frac{\sigma}{40 \times 10^9}$

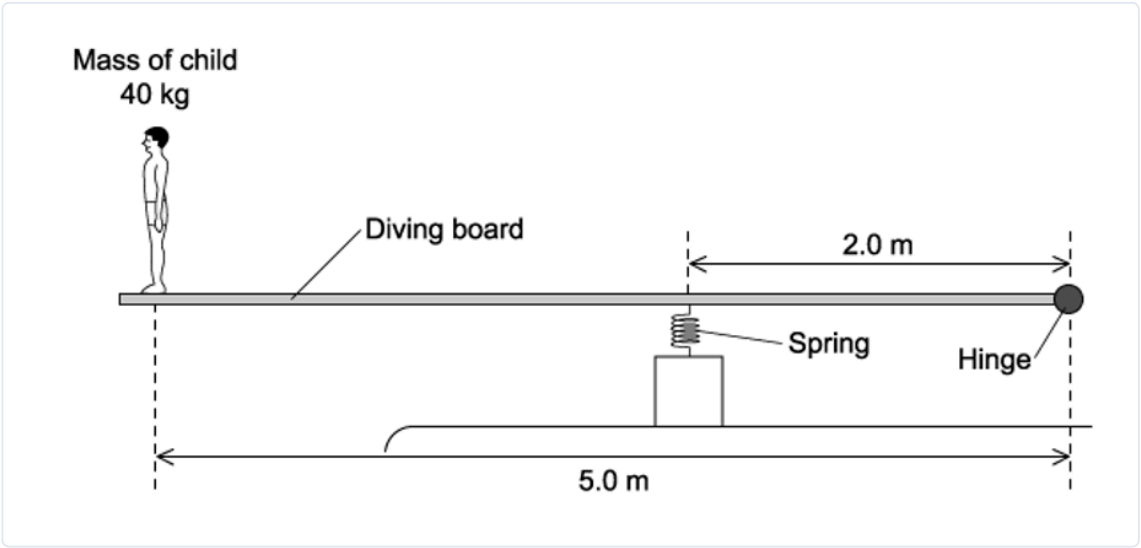
- $E_n = \sigma \times \frac{1.0}{2 \times 10^9} = \frac{\sigma}{2 \times 10^9}$

- The greatest stress that can be added is the sum of the stress of both materials, and this should be equal to 3.0 mm
  - $E_n + E_P = 3.0 \times 10^{-3}$
  - $\frac{\sigma}{2 \times 10^9} + \frac{\sigma}{40 \times 10^9} = 3.0 \times 10^{-3}$
  - $\sigma \times \frac{1}{2 \times 10^9} + \frac{1}{40 \times 10^9} = 3.0 \times 10^{-3}$
  - $\sigma = 3.0 \times 10^{-3} \quad / \frac{1}{2 \times 10^9} + \frac{1}{40 \times 10^9} = 5.71 \times 10^6 \quad \text{Pa}$

- Therefore, the greatest tensile stress that can be applied is$5.7 \times 10^6 \quad \text{Pa}$

Question 21 Hard [1 mark]

A child with a mass of 40 kg was standing on a diving board. The diving board has a length of 5.0 m and is hinged at one end then supported by a spring 2.0 m from the hinged end. The spring has a spring constant of 10 kN m^{-1}



What is the extra compression of the spring caused by the child standing on the end of the board?

A 16 cm

B 9.8 cm

C 1.6 cm

D 1.0 cm

Mark Scheme & Explanation:

*The correct answer is **B** because:*

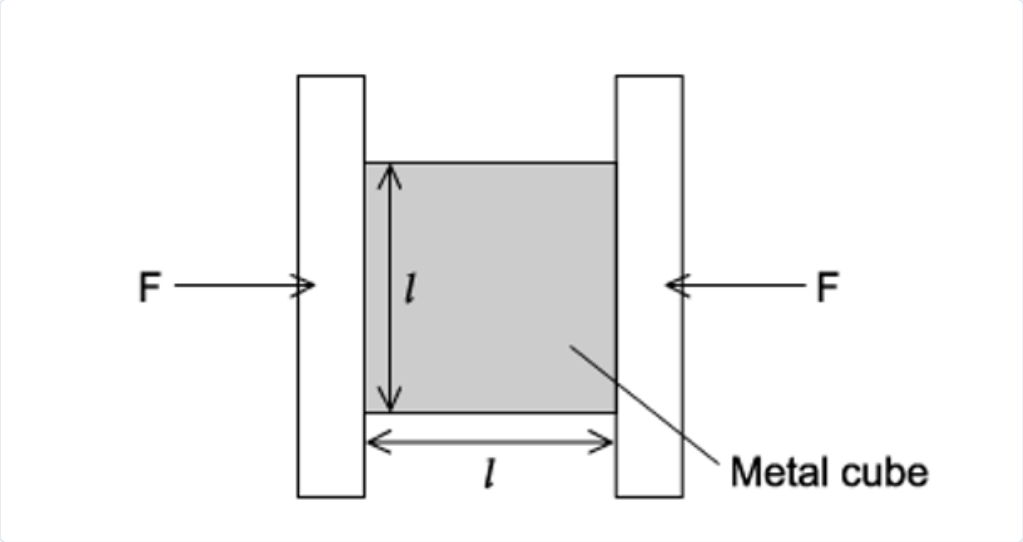
- Hooke's law states $F = kx$, where *k* is the spring constant
- The child and the spring will cause a moment at a distance from the hinge
- At equilibrium, the torque of the child (anticlockwise) should be equal to the torque of the spring (clockwise)
- Let the force of the spring = *F*
  - Clockwise moments $= mgx = 40 \times 9.81 \times 5.0$
  - Anticlockwise moments $= Fd = F \times 2.0$
  - $40 \times 9.81 \times 5.0 = F \times 2.0$

- $F = \frac{40 \times 9.81 \times 5}{2} = 981 \quad \text{N}$

- So, the spring force is equal to F = 981 N
- Spring constant, k = 10 kN m^{-1} = 10000 N m^{-1}
  - Extension $= \frac{F}{k} = \frac{981}{10000} = 0.0981 \quad \text{m}$
- Therefore, the compression of the spring caused by the child is 9.8 cm

Question 22 Hard [1 mark]

The diagram shows a metal cube is placed between two board and compressed elastically by two opposing forces *F*. The cube has a length of each side of l



How will Δ*l*, the amount of compression, relate to *l* ?

A $\Delta l \ \alpha \ l$

B $\Delta l \ \alpha \ l^2$

C $\Delta l \ \alpha \ \frac{1}{l^2}$

D $\Delta l \ \alpha \ \frac{1}{l}$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The Young modulus is given by $E = \frac{stress}{strain} = \frac{\sigma}{\varepsilon}$
  - Stress is equal to $\sigma = \frac{force}{area} = \frac{F}{A}$
  - Strain is equal to $\varepsilon = \frac{extension}{original \ \ length} = \frac{\Delta L}{L}$
- Therefore, Young modulus $E = \frac{\frac{F}{A}}{\frac{e}{L}} = \frac{FL}{Ae}$
- Where:
  - Area $A = l^2$

    - Compression $e = \Delta l$
- So, Young modulus is proportional to:
    - $E \propto \frac{l}{l^2 \times \Delta l}$
    - $E \propto \frac{1}{l \times \Delta l}$
    - So $\Delta l \propto \frac{1}{El}$, giving $\Delta l \propto \frac{1}{l}$ as the Young modulus is constant

Question 23 Hard [1 mark]

A metal wire that is supported vertically from a fixed point has a load of 84 N applied to the lower end. The wire has a cross-sectional area of 0.02 mm^2 and obeys Hooke's law. The length of the wire increases by 0.30 %.

What is the Young modulus of the metal wire?

A 1.4×10^5 Pa

B 1.4×10^8 Pa

C 1.4×10^9 Pa

D 1.4×10^{12} Pa

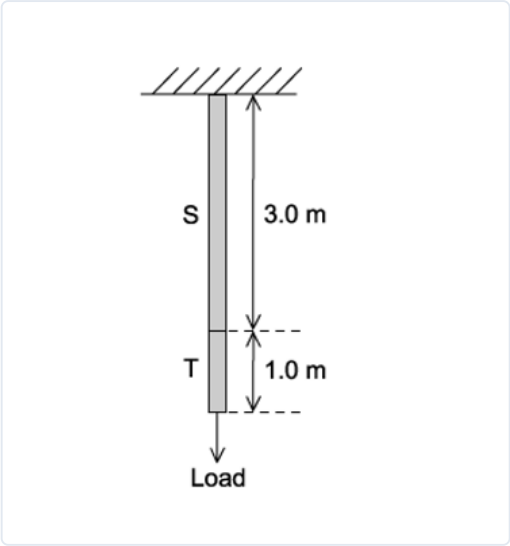
Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The Young modulus is given by $E = \frac{stress}{strain} = \frac{\sigma}{\varepsilon}$
  - Stress is equal to $\sigma = \frac{force}{area} = \frac{F}{A}$
  - Strain is equal to $\varepsilon = \frac{extension}{original\ \ length} = \frac{\Delta L}{L}$
- Where:
  - Force F = 84 N
  - Area $A = 0.02 \text{ mm}^2 = 0.02 \times 10^{-6} \text{ m}^2$
- The wire increases in length by 0.3 %, this is equal to the strain, so:
  - $Strain = 0.3\% = \frac{0.3}{100} = 0.0030$
- Young modulus = stress / strain
  - $E = \frac{84}{0.02 \times 10^{-6} \times 0.003} = 1.4 \times 10^{12}$ Pa
- Therefore, the Young modulus of the metal wire is 1.4 × 10^{12} Pa

Question 24 Hard [1 mark]

Two wires S and T is joined, S has a length of 3.0 m and T a length of 1.0 m, both wires have a uniform cross-sectional area.



When a load was hung from the wires S stretched by 1.5 mm and T by 1.0 mm.

The same load is then hung from a second wire of the same cross-section, consisting of 1.0 m of metal S and 3.0 m of metal T.

What is the total extension of this second wire?

A 3.5 mm

B 2.5 mm

C 5.0 mm

D 4.8 mm

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The Young modulus is given by $E = \frac{stress}{strain} = \frac{\sigma}{\epsilon}$
  - Stress is equal to $\sigma = \frac{force}{area} = \frac{F}{A}$
  - Strain is equal to $\varepsilon = \frac{extension}{original\ \ length} = \frac{\Delta L}{L}$
- In a material the Young modulus remains constant

- As the load and cross-sectional area are the same, the stress $\frac{force}{area}$ is going to be the same
- Metal S
    - 1st experiment, $E = \frac{3}{1.5}$
    - 2nd experiment $E = \frac{1}{x}$
    - Therefore, $\frac{3}{1.5} = \frac{stress}{x}$
    - Extension, $x = \frac{1.5}{3} = 0.5 \quad \text{mm}$
- Metal T
    - 1st experiment, $E = \frac{1}{1}$
    - 2nd experiment $E = \frac{3}{x}$
    - Therefore, stress is equal to $= \frac{3}{x}$
    - So, extension $x = 3.0 \quad \text{mm}$
- Total extension $0.5 + 3.0 = 3.5 \quad \text{mm}$

Question 25 Hard [1 mark]

Two steel wires X and Y that both obey Hooke's law were used in an experiment. Wire X has a length l and a cross-sectional area A. Wire Y has a length of $2l$ and cross-sectional area of $\frac{A}{2}$.

What is the ratio $\frac{tension\ \ in\ \ X}{tension\ \ in\ \ Y}$ when both wires are stretched to the same extension?

A $\frac{1}{4}$

B $\frac{4}{1}$

C $\frac{2}{1}$

D $\frac{1}{2}$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The Young modulus is given by $E = \frac{stress}{strain} = \frac{\sigma}{\epsilon}$
  - Stress is equal to $\sigma = \frac{force}{area} = \frac{F}{A}$
  - Strain is equal to $\varepsilon = \frac{extension}{original\ \ length} = \frac{x}{L}$
- As both wires are made of steel they have the same Young modulus, *E*
- Young modulus $E = \frac{\frac{F}{A}}{\frac{x}{L}} = \frac{FL}{Ax}$
- So, tension is equal to $F = \frac{EAx}{l}$
  - For wire X : $F_X = \frac{EAx}{l}$
  - For wire Y : $F_Y = \frac{E\frac{A}{2}x}{2l} = \frac{EAx}{4l} = \frac{F_X}{4}$
- So, the ratio $= \frac{F_X}{F_Y} = \frac{F_X}{\frac{F_X}{4}} = \frac{4}{1}$